Tropical Cyclone Wind Intensity Drives Phase-Specific Displacement of Emergency Department Demand in Florida, U.S., 2004–2022

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Abstract

Tropical cyclones (TCs) impose substantial public health burdens, yet day-to-day shifts in emergency department (ED) utilization before, during, and after storm landfall remain imprecisely characterized. Using 19 years of ZIP-code-level data from Florida, U.S., we estimated expected ED visits surrounding TC events. The Threat phase was defined as the eight days immediately preceding local storm impact, corresponding to forecast approach and heightened anticipatory response; the Impact phase as the single day of closest storm approach or landfall in Florida, when hazardous conditions are most acute; and the Cleanup phase as the eight days immediately following impact, reflecting the early recovery period marked by infrastructure disruption and preliminary recovery activities. We further examined how these patterns scaled with storm intensity (≥34 [n = 44], ≥55 [n = 23], and ≥64 kt [n = 14]). Using a two-part model with ZIP, year-month, and day-of-week fixed effects, including a population offset in the count component and population-weighted averaging, we estimated phase-specific differences in expected ED visits relative to non-storm conditions. Across all qualifying TCs (≥34 kt), Threat and Cleanup phases were associated with increases of 0.32 and 0.34 expected ED visits per ZIP-day, respectively, whereas the Impact phase was associated with a decrease of 1.45 expected visits per ZIP-day.

These effects intensified with storm strength: Impact phase reductions deepened from −1.45 expected visits per ZIP-day at ≥34 kt to −2.78 at ≥55 kt and −3.72 at ≥64 kt (hurricane force), while Cleanup phase increases rose from +0.34 to +0.61 and +1.01 expected visits per ZIP-day. These findings suggest a phase-specific displacement–disruption pattern in ED utilization, with implications for anticipatory staffing, access planning, and post-storm surge capacity in Florida.

Keywords: Tropical cyclones; healthcare; emergency department visits; climate change; Florida; wind intensity

1. Introduction

Tropical cyclone (TC) related health burdens extend beyond immediate landfall, altering access to healthcare, evacuation and displacement, infrastructure and transport disruptions, and other hazards worldwide [1]. TC-related health impacts arise through both direct and indirect mechanisms. In Florida, U.S., direct and indirect mortality rates were elevated in the months following TCs Charley, Frances, Ivan, and Jeanne in 2004 [2]. Direct trauma upon landfall accounted for only 4% of total fatalities, while indirect mortality, predominantly cancer and cardiovascular-related deaths, persisted for up to two months after each storm [2]. Evacuation procedures prior to TC landfall have also been shown to carry health risks; among Florida assisted living residents exposed to Hurricane Irma, evacuation was associated with 16% greater odds of emergency department visits within 30 days compared to those who sheltered in place [3].

Studies examining the relationship between TC wind speed intensity and adverse health outcomes have documented dose-response patterns, suggesting that health burden scales systematically with wind speed intensity. However, these analyses are primarily through mortality and hospitalization counts aggregated over post-storm windows at the global, state, or county level. For example, in a large-scale analysis across the contiguous United States for TCs between 1930

and 2015, Young and Hsiang [4] found that the average TC can generate an estimated 7,000–11,000 excess deaths that far exceed the official fatality counts, where the nonlinear relationship between wind intensity and all-cause mortality was found to persist for up to 15 years following a storm. In a global study spanning nine countries linking TC maximum sustained winds and cause-specific mortality, a linear relationship was found to persist with maximum sustained wind speeds beyond 68 kt [1].

Despite this evidence linking TCs to public health burdens in Florida, most empirical studies of TC health impacts define exposure windows beginning at or after landfall, leaving the pre-impact and phase-specific structure of ED demand largely uncharacterized. County-level analyses have documented significant increases in cause-specific mortality [5] and hospitalizations across multiple disease categories following TC exposure [6], and longer analyses suggest these effects can persist for months to years after TC exposure across the full range of storm intensities, with stronger storms producing larger and more prolonged mortality burdens [4]. However, many of these previous studies aggregate county-exposed storm-days into a specific post-event window, with little differentiation between the threat, impact, and recovery phases. Reports on storm-related hospitalization and mortality counts are normally anchored on the impact day alone, and any pre-landfall coefficients that may be included are rarely interpreted in a meaningful way [7, 8].

Compared to ZIP codes, county-level analyses can contain more exposure misclassification errors by averaging over impacted and non-impacted areas. Florida counties can span rural and urban areas, where population densities can vary from coastal metropolitan to sparsely settled inland communities. Spatial resolution at the ZIP code level enables more accurate alignment between TC exposure, particularly in urban and suburban areas, improving local control of baseline variation in ED utilization.

More recently, evidence indicates that TC-related health risks in Florida are temporally distributed and may begin prior to landfall, with measurable increases in all-cause mortality during the pre-impact threat period that intensify through impact and into recovery [9]. Despite this body of evidence linking TCs to public health burdens in Florida, the daily phase-specific impacts of TCs on emergency department (ED) utilization have not been well characterized. In this study, we quantify ED utilization changes across distinct phases of a tropical cyclone event, including the days preceding landfall, the day of impact, and the extended recovery period, and assess whether these changes vary by storm intensity. We integrate 19 years of Florida ED data with spatially explicit storm swaths to estimate phase-specific daily changes in ED visit counts at the ZIP code level, across three storm intensity thresholds.

2. Methods

2.1 Study location

Florida is uniquely suited for this analysis beyond data availability. It has one of the largest and most diverse state populations in the US, with over 23 million residents concentrated along approximately 1,350 miles of general coastline exposed to TC landfalls from both the Atlantic and Gulf of Mexico. Florida has recorded more hurricane landfalls than any other state, accounting for approximately 40–41% of all US hurricane landfalls since 1851 [10, 11]. Its low elevation peninsular geometry means it has limited upstream land to dissipate storm intensity upon TC landfall [12, 13].

2.2 Storm swath construction and phase classification

We constructed storm swaths from the International Best Track Archive for Climate Stewardship (IBTrACS) tracks based on the radius of maximum wind for each TC (≥34 kt

≈17 m/s) impacting Florida over the period 2004–2022 (n = 44 storms). Next, we labeled ZIP-code-level exposure within ±8 days of landfall as Threat (n = 64,240 ZIP-days), Impact (n = 8,030), or Cleanup (n = 64,230), resolving overlapping storm effects (rare) by privileging Impact over Threat and Threat over Cleanup to build a ZIP-by-date effect calendar linked to daily ED counts. ZIP-days falling outside any storm window were classified as None and served as the reference baseline against which phase-specific effects were estimated (n = 6,740,950 ZIP-days). Thirty of the 44 storms over the period were tropical storm strength, with the remaining fourteen broken down by Saffir-Simpson category as follows: three Category 1, three Category 2, four Category 3, two Category 4, and two Category 5. Over the 19-year period, 2005 had the most storms with six and 2014 and 2015 had no qualifying storms.

2.3 Emergency department data and data suppression

The Florida Department of Health Agency for Healthcare Administration and Environmental Public Health Tracking Network shared the ED encounter data (Florida State University's Institutional Review Board #5297, Florida Department of Health Institutional Review Board #2018-066-FSU). To protect privacy, the data provider suppressed daily ED counts of 6 or fewer, so a two-part fixed-effects model was used to accommodate suppression of low daily counts. Counts of six or fewer were reported at the suppression threshold; therefore, we treated records with value 6 as representing ZIP-days with true counts ≤6. The first part modeled the probability that a ZIP-day count was at or below the suppression threshold using logistic regression. The second part modeled expected ED counts among ZIP-days with observed counts greater than six using Poisson regression. In both parts, we included ZIP code, year-month, and day-of-week for statistical control. We translated model coefficients into marginal, phase-specific expected ED visits by counterfactually assigning each ZIP-day to a given storm phase and

averaging predictions from the two-part model, yielding contrasts versus the None condition. Phase effects were estimated separately within each storm-intensity stratum (≥34 [n = 44], ≥55 [n = 23], and ≥64 kt [n = 14]), representing tropical-storm-force winds, a stronger tropical-storm threshold, and hurricane-force winds, respectively, and interpreted jointly to assess intensity scaling.

2.4 Population weighting

To account for differences in ZIP-code population size, we incorporated annual ACS 5-year ZCTA population estimates, interpolated available years to a complete annual series, and carried the nearest available estimate for years outside the ACS range before expanding annual values to monthly ZIP-level population values [14]. We treated ZCTAs as equivalent to ZIP codes. In the Poisson component of the two-part model, we included the logarithm of ZIP-month population as an offset, so that expected counts were adjusted for the size of the population at risk. The predicted expected ED visits were averaged using ZIP-month population weights. Thus, the primary target variable is a population-weighted difference in expected ED visits per ZIP-day under each counterfactual storm phase relative to the None condition.

2.5 Falsification analyses

As falsification analyses, we repeated timing-randomization procedures using the same fixed-effects two-part model structure and examined whether randomized storm timing reproduced the direction and magnitude of the main phase-specific pattern. For each storm swath, the observed impact date was replaced with a randomly assigned date within the same hurricane season (May–November) and calendar year, constrained to be at least 16 days from the true impact date. Using these randomized impact dates, we reconstructed the ZIP×date effect calendar with Threat, Impact,

and Cleanup phases defined over ±8 days, resolved overlaps by severity, and re-estimated the same two-part model specification used in the main analysis (ZIP, year-month, and day-of-week fixed effects). We repeated this procedure 200 times and summarized the distribution of estimated coefficients across iterations. As an additional falsification analysis, we conducted a placebo test that preserves the empirical distribution of storm-phase labels within each ZIP code and calendar month while breaking their temporal alignment with storm events. Specifically, within each ZIP×month block, storm-phase labels (None, Threat, Impact, Cleanup) were randomly permuted across days, holding fixed the marginal frequency of each phase and all outcome data. This procedure was also repeated 200 times, and the distribution of estimated phase coefficients was summarized across iterations.

3. Results

3.1 Phase-resolved combined effects

We begin by considering all qualifying TCs with maximum sustained winds ≥34 kt, including tropical storms and hurricanes. Averaging modeled predictions over ever-exposed ZIP codes with ZIP-month population weights, expected visits under the None condition averaged 17.50 per ZIP-day (95% CI: 16.8–18.1). Relative to this baseline, the Threat phase was associated with +0.32 expected visits per ZIP-day (95% CI: 0.27–0.38), the Impact phase with −1.45 expected visits per ZIP-day (95% CI: −1.58 to −1.29), and the Cleanup phase with +0.34 expected visits per ZIP-day (95% CI: 0.28–0.41).

3.2 Intensity stratification

When the analyses were repeated for higher wind-intensity subsets, effects increased systematically with wind speed (figure 1). The Impact phase reductions were −1.45 expected visits

per ZIP-day (95% CI: −1.58 – −1.29) at ≥34 kt, −2.78 expected visits per ZIP-day (95% CI: −3.07–−2.53) at ≥55 kt, and −3.72 expected visits per ZIP-day (95% CI: −4.07–−3.39) at ≥64 kt. Increases observed during the Cleanup phase scaled correspondingly from +0.34 expected visits per ZIP-day (95% CI: 0.28–0.41) to +0.61 expected visits per ZIP-day (95% CI: 0.44–0.76) and +1.01 expected visits per ZIP-day (95% CI: 0.81–1.22). Threat phase effects remained modest and positive across all thresholds (+0.32, +0.28, and +0.50, respectively), without a consistent intensity-scaling pattern. As shown in figure 1, the monotonic intensification of both Impact suppression and Cleanup rebound with increasing storm strength is visually apparent across the three intensity panels, while Threat estimates cluster near zero with overlapping confidence intervals across the wind intensity categories.

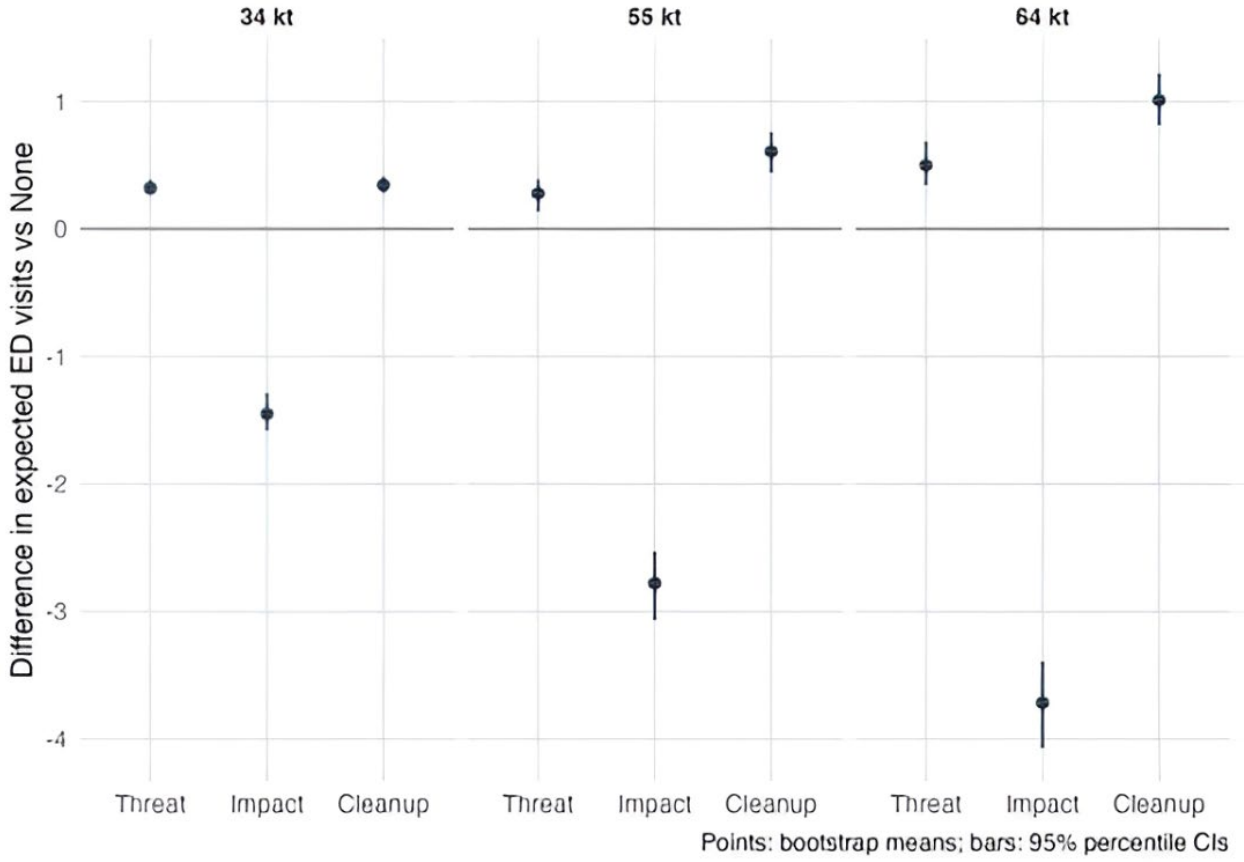


Figure 1. *Population-weighted differences in expected emergency department (ED) visits per ZIP-day relative to non-storm conditions, by storm phase and storm-intensity threshold. Each panel corresponds to a storm-intensity threshold (≥34, ≥55, and ≥64 kt). Estimates were obtained from a two-part model with ZIP, year-month, and day-of-week fixed effects; the Poisson count component included a ZIP-month population offset, and counterfactual predictions were averaged using ZIP-month population weights. Dots show point estimates, and error bars show 95% confidence intervals from 200 ZIP-clustered bootstrap resamples.*

3.3 Placebo tests

Across 200 placebo iterations, estimated phase effects were small in magnitude and centered near zero. In the logistic component (probability of ≤6 ED visits), mean placebo coefficients were −0.020 (Cleanup), −0.018 (Impact), and −0.016 (Threat), with narrow confidence intervals reflecting sampling variability rather than systematic effects. In the Poisson component (visits > 6), mean placebo coefficients were close to zero for all phases (absolute values ≤0.001), with confidence intervals spanning or tightly bracketing zero. No consistent displacement–disruption pattern emerged under randomization, supporting the interpretation that the observed phase-specific effects in the main analysis reflect storm timing and exposure rather than seasonal confounding or model artifacts.

Additionally, across 200 ZIP×month randomizations, estimated phase coefficients were again small in magnitude. In the logistic component, mean placebo coefficients ranged from −0.026 to −0.029 across phases, while corresponding Poisson coefficients were positive but near zero (0.003–0.004). Although some mean estimates differed slightly from zero, their magnitudes were orders of magnitude smaller than the effects observed in the true storm-timing analysis and did not reproduce the characteristic Threat increase, Impact suppression, and Cleanup rebound pattern. Together with the randomization-in-time placebo, these results indicate that the main

findings are unlikely to be driven by residual seasonality, within-month utilization patterns, or mechanical features of the model specification.

4. Discussion

ED utilization across Florida ZIP codes follows a consistent displacement–disruption pattern in response to TCs: modest pre-storm increases during the Threat phase, sharp decreases on the Impact phase, and post-storm rebounds during the Cleanup phase. The pattern strengthens monotonically with increasing wind intensity for the Impact and Cleanup phases. In addition, by characterizing daily phase-specific ED changes at the ZIP-code level, we emphasize the understudied impacts of TCs during the Threat phase and suggest that more research is needed to understand these increases. The sharp decreases of ED utilization on the Impact phase potentially reflect a combination of reduced mobility, healthcare facility closures, and shelter-in-place behavior rather than reductions in medical need [6, 15, 3]. As a result, deferred medical care accumulates during the Impact phase and re-enters the healthcare system during the Cleanup phase, producing the observed increases after TC landfall. The increased ED visits during the Threat phase may also indicate that populations increase health-seeking behavior (evacuation-related and stress-related care) ahead of TC landfall, which may require an additional preparedness window for health systems to manage demand for care before storm conditions deteriorate.

The ED visit suppression observed during the Impact phase in our results agrees with previous studies documenting decreases in hospitalization rates on TC exposure days. Similarly, the rebound in visits during the Cleanup phase observed here is consistent with the body of literature. However, while our findings directionally align with prior studies that found a dose-response pattern [6, 1], we observe greater magnitude of the effects relative to wind intensity

scaling. The reasons are likely rooted in the spatial resolution of our data at the daily ZIP code level. The dose-response pattern in our model strengthens with more intense wind speeds, where Impact phase ED suppression in population-weighted expected visits per ZIP-day nearly triples from −1.45 at ≥34 kt to −3.72 at ≥64 kt (figure 1), and the rebound during the Cleanup phase more than doubles across the same intensity range. The ZIP-days that contribute most to our estimates are ZIP codes where daily visit counts are generally higher, which tend to be urban and suburban ZIP codes. To this point, the areas with higher TC exposure risk are those along the Atlantic and Gulf coasts of the peninsula, which are disproportionately urban ZIP codes. Therefore, our results preserve the population density heterogeneity between urban and rural ZIP codes that enable the sharper wind category signals to emerge.

Prior evidence suggests that short-term behavioral responses to impending TCs may increase emergency care-seeking before landfall, even in the absence of acute storm hazards. Evacuation planning, relocation, and anticipatory disruptions to daily routines can prompt precautionary or problem-driven ED visits, particularly among medically complex populations. For example, evacuation of long-term care and assisted living residents has been linked to increased emergency department use, reflecting challenges in care continuity, medication access, and symptom management during storm preparation rather than direct injury [16, 3, 17]. Anticipated pharmacy closures, transportation barriers, or uncertainty about access to outpatient care may further encourage individuals to seek emergency services earlier than they otherwise would. Heightened psychological stress during the forecast and warning period, driven by evacuation decisions, prior storm experiences, and intensive media coverage, may also increase ED presentations for anxiety-related, cardiovascular, or respiratory complaints [18, 19]. Temporary disruptions in medication adherence or supply may similarly lead patients to seek

emergency care for symptom destabilization within days of TC impact. Together, these mechanisms are consistent with a pre-impact increase in ED utilization driven by anticipatory and precautionary care-seeking behavior, rather than by direct storm exposure.

We note that our model construction requires some considerations when interpreting results. First, storm exposure is approximated using the radius to maximum wind from IBTrACS, which captures the area of peak wind intensity but may not fully represent the spatial extent and regional heterogeneity of hazardous conditions, particularly for rainfall and storm surge. Storm variables can vary based on geography; future research can use parametric surface wind models to estimate local surface wind levels. Second, storm exposure does not align precisely with forecast threat areas, especially forecast lead times greater than four days. Third, the data suppression threshold of 6 or fewer visits per ZIP-day, while addressed by the two-part model, introduces uncertainty in the lower tail of the visit distribution. Fourth, population displacement from evacuation may affect the number of exposed residents; however, there is no reliable dataset on short-term evacuations over the study period. Consequently, the study's ZIP-month population denominators may overstate the population physically present in evacuated areas during the day of impact. Nevertheless, the consistency of our results across wind intensity categories and the robustness of the fixed-effects design to ZIP code level show that ED visits associated with TCs have a consistent pattern in Florida. Future additions to this analysis include exploring how phase-specific ED visits compare at the weekly-county level. Weekly aggregation would provide large enough counts to avoid data suppression rules, facilitating the examination of cause-specific ED visits across many disease categories and demographics. County weekly-level aggregation preserves rural area population samples more effectively, allowing for a comparison to ZIP code aggregation across the wind intensity categories.

5. Conclusions

Over a 19-year period, we characterized phase-specific ED utilization associated with TCs at the ZIP-code level in Florida. Combining output from two fixed-effects models, we found a robust ED utilization pattern consisting of modest increases during the Threat and Cleanup phases and a sharp decrease during the Impact phase. The estimated decreases in expected ED visits during the Impact phase and the increases during the Cleanup phase both strengthened as storm intensity increased from 34 to 55 to 64 knots, while increases during the Threat phase remained modest and did not follow a consistent intensity-scaling pattern. However, the modest increases during the Threat phase have not previously been documented at the ZIP-code level across a multi-decade record and add an important dimension to our understanding of TC impacts on Florida communities. These findings have direct implications for health system preparedness. The sharp decreases in ED utilization at TC landfall likely reflect a combination of mobility constraints due to deteriorating conditions, facility closures, and shelter-in-place procedures that likely defer medical care from the Impact phase to the Cleanup phase.

More research is needed to understand the drivers leading to the increases in ED utilization during the Threat phase. The consistency of results across intensity categories and the robustness of the fixed-effects design provide confidence in the pattern of ED visits identified here. As coastal populations grow and as TC intensity is projected to increase with anthropogenic climate change, phase-specific and intensity-scaled ED utilization estimates of the kind developed here will become increasingly useful tools for emergency preparedness planning.

Acknowledgments

Any published findings and conclusions are those of the authors and do not necessarily represent the official position of the Florida Department of Health. The research was supported by the National Academies of Science Gulf Research Program Understanding the Effects of Climate Change on Environmental Hazards in Overburdened Communities (SCON-10000677). The authors declare no competing interests. All analysis code (including storm swath construction, effect calendar, two models, and the bootstrap routines) are available in a public repository (github.com/jelsner/ER-Visits). ED microdata is subject to data-use agreements; aggregated products and synthetic examples will be shared as permitted.

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